



PRANALI SANDEEP HAGARE

AI & Data Science Engineer

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🌐 Portfolio · 🔗 LinkedIn · 🐙 GitHub

Professional Summary

Results-driven **AI & Data Science Engineer** with expertise in Python, Machine Learning, Deep Learning, SQL, and Generative AI (RAG, LangGraph, LLMs). Proven track record in building end-to-end data pipelines, real-time computer vision translation systems, and agentic AI platforms. Co-author of an Indian Govt. Copyright and an IEEE publication. Skilled in translating complex datasets into high-accuracy ML/DL solutions.

Technical Skills

Languages & Core: Python, SQL, Object-Oriented Programming (OOP)

AI / Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, EDA, Regression, Classification, Clustering, Random Forest

Deep Learning & Neural Networks: ANN, CNN, RNN, LSTM, Bidirectional GRU (Bi-GRU)

Generative AI & LLMs: RAG (Retrieval-Augmented Generation), LangGraph, Groq API, Vector Databases (ChromaDB), Embeddings (sentence-transformers), Prompt Engineering

Web & Backend: FastAPI, Flask, REST APIs

Databases & DevOps: PostgreSQL, Supabase, Git, GitHub Actions (CI/CD)

Work Experience

Test Yantra

Nov 2025 – Apr 2026

Associate Software Engineer

On-site

- Engineered Python backend APIs and database integration modules for enterprise application workflows.
- Integrated **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** into backend systems to automate query handling and data extraction.

Oasis Infobyte

Jul 2024 – Dec 2024

Machine Learning Intern

Remote

- Developed, trained, and evaluated supervised and unsupervised machine learning algorithms using **Python** on real-world datasets.
- Conducted Exploratory Data Analysis (EDA), feature selection, and data preprocessing with **Pandas** and **NumPy** for predictive modeling pipelines.

Projects

HireSense – AI Resume Intelligence Platform | FastAPI, LangGraph, RAG, ChromaDB, Groq

2026

- Architected a full-stack AI talent platform featuring candidate and recruiter portals for automated ATS resume auditing and match scoring.
- Built multi-step agentic workflows using **LangGraph** and **Groq (llama-3.3-70b)** with vector search via **ChromaDB** and **sentence-transformers**.

SignBridge AI – Real-Time Sign Language Interpreter | MediaPipe, OpenCV, Bi-GRU, FastAPI

2024

- Developed a real-time gesture translation engine converting ISL/ASL signs into text and natural speech at **60 FPS** with **96.25% accuracy**.
- Formulated ambidextrous landmark feature pipelines across 50+ video sequences per gesture using a **Bidirectional GRU / LSTM** architecture.

CropSense – Precision Agriculture Optimization Engine | Python, Flask, GitHub Actions

2024

- Trained a **Random Forest** classifier (100 trees) on 2,200 soil/climate records, achieving **99.32% test accuracy** across 22 crop categories.
- Deployed a **Flask REST API** backend with a single-page interactive UI, automating deployment via **GitHub Actions CI/CD**.

Education

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering & Technology

June 2024

Bachelor of Engineering (B.E.) in Artificial Intelligence & Data Science – CGPA: 8.19 / 10.0

Baramati, Pune

Publications & Patents

Government of India Copyright: “Empowering Deaf by Translating Emergency Signs in Indian Sign Language to Text”, Copyright Office, Govt. of India. Reg No: L-137992/2023.

IEEE International Conference Paper: “Empowering Deaf with Indian Sign Language Interpreter using Deep Learning”, 2024 IEEE MITADTSOCiCon, Pune, India, pp. 1-6, DOI: 10.1109/MITADTSOCiCon60330.2024.10575064.